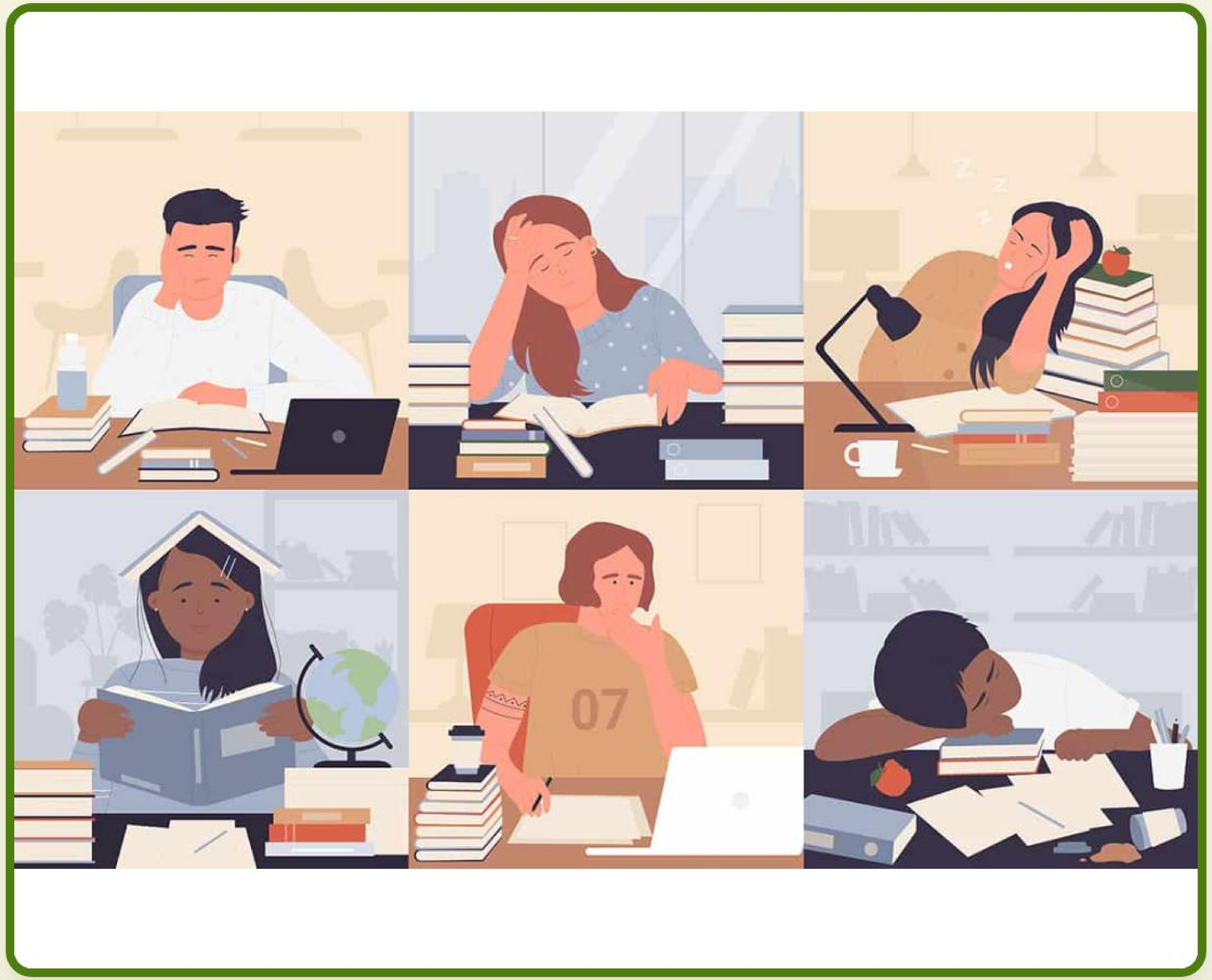
From Messy Excel to AI-Ready Data

A Data Architecture Workshop for Plant Scientists

The Trap of the "Pretty" Spreadsheet

- As plant scientists, you collect incredible amounts of data in the field and lab.
- We naturally format Excel to look good to human eyes: merged cells, blank spacer rows, and color-coding.
- **The Problem:** Software (like SPSS, R, Python) and AI tools look at *structure*, not visuals.
- Making data "pretty for humans" often breaks it for computers, requiring hours of manual cleaning before analysis can begin.



The Three Golden Rules of Tidy Data

How to bridge the gap between human observation and machine analysis.

Rule 1: One Observation = One Row

The Bad Habit

Leaving cells blank to visually group items. For example, typing "Plant 1" in row 1, and leaving the next 9 rows blank because they are all leaves belonging to Plant 1.

When an AI reads row 5, it sees a blank ID. If you sort or filter this data, the leaf data is completely detached from its source plant.

The AI-Ready Fix

Repeat the identifier on **every single row**. If you measure 10 leaves from Plant 1, the text "Plant 1" must appear 10 times in the ID column.

Pro-tip: Just double-click the bottom right corner of the Excel cell to auto-fill instantly.

Example: One Observation = One Row

✕ Messy Data (Implicit)

Location	Plant ID	Leaf Number	Leaf Length (cm)
Pahang	Wild-01	Leaf 1	5.2
Blank	*Blank*	Leaf 2	5.8
Blank	*Blank*	Leaf 3	4.9

✓ AI-Ready Data (Explicit)

Location	Plant ID	Leaf Number	Leaf Length (cm)
Pahang	Wild-01	Leaf 1	5.2
Pahang	Wild-01	Leaf 2	5.8
Pahang	Wild-01	Leaf 3	4.9

Rule 2: One Variable = One Column

- **The Bad Habit:** Creating merged headers. For instance, a giant merged cell saying "Stem" on top, with "Height" and "Diameter" underneath it in a second row.
- **The Problem:** Statistical software and AI expect exactly one row of variable names. Stacked headers cause tools to misinterpret column names or fail to load the data entirely.
- **The AI-Ready Fix:** Flatten your headers into exactly one row. Combine concepts using underscores, such as `Stem_Height` and `Stem_Diameter`.

Example: One Variable = One Column

✕ Messy Data (Stacked / Merged Headers)

Stem Measurements (Merged)		Flower Color
Height (cm)	Diameter (cm)	
15.2	2.1	Yellow
14.5	1.9	Yellow

✓ AI-Ready Data (Flat Headers)

Stem_Height_cm	Stem_Diameter_cm	Flower_Color
15.2	2.1	Yellow
14.5	1.9	Yellow

Rule 3: One Value = One Cell

The Bad Habit

Typing ranges like "7.0 - 9.6" for leaf length, or entering text like "N/A - plant died" in a numeric column.

The moment you add a dash or text to a column of numbers, Excel converts the whole column to "Text". You instantly lose the ability to calculate averages, minimums, or graph the data.

The AI-Ready Fix

Record the exact raw numbers in different rows, and let Excel or AI calculate the summary range later.

If data is missing, just leave the cell **completely blank**. Pick one consistent rule for missing data and stick to it.

Example: One Value = One Cell

✕ Messy Data (Mixed Types)

Plant ID	Leaf Length (cm)	Chlorophyll
Wild-01	**7.0 - 9.6**	45.2
Wild-02	8.1	**- (Broken)**

✓ AI-Ready Data (Raw Numbers)

Plant_ID	Leaf_Length_cm	Chlorophyll	Notes
Wild-01	7.0	45.2	
Wild-01	9.6	45.5	
Wild-02	8.1		Broken

Bonus Mistake 1: Colors are NOT Data

✕ Messy Data (Using Cell Highlight)

AI and statistical tools cannot "see" Excel background colors. This data is lost entirely.

Plant ID	Location	Stem Height (cm)
Wild-01	Pahang	15.2
Wild-02	Pahang	14.5

✓ AI-Ready Data (Explicit Columns)

Plant_ID	Location	Stem_Height_cm	Health_Status
Wild-01	Pahang	15.2	Healthy
Wild-02	Pahang	14.5	**Diseased**

Bonus Mistake 2: Units Belong in Headers

✕ Messy Data (Units inside Cells)

Writing "cm" inside the cell turns your number into a string of text. You can't calculate a mean of "15 cm".

Plant ID	Root Length	Biomass
Cultivated-A	**12 cm**	**14.2 g**
Cultivated-B	**14 cm**	**16.1 g**

✓ AI-Ready Data (Units in Header)

Plant_ID	Root_Length_cm	Biomass_g
Cultivated-A	12.0	14.2
Cultivated-B	14.0	16.1

Bonus Mistake 3: Don't Stuff Cells

✕ Messy Data (Multiple Variables in One Cell)

Combining categories makes it impossible to quickly filter or count how many plants had aphids.

Plant ID	Pests Observed
Wild-03	**Aphids, Spider Mites**
Wild-04	**Whiteflies**

✓ AI-Ready Data (Split Categorical Variables)

Plant_ID	Has_Aphids	Has_Spider_Mites	Has_Whiteflies
Wild-03	Yes	Yes	No
Wild-04	No	No	Yes

Interactive Exercise

Spot the mistakes in the next slide's dataset.

Spot the Mistakes: What will break the AI?

Location	Plant ID	Stem (Merged)		Leaf Length (cm)	Note
		Height	Width		
Pahang	Wild-01	15.2	2.1	5.5	
				6.1	
				5.8 - 6.0	
	Wild-02	14.5	-	4.2	Dead

Answer Key: The Fixes

- **Stacked/Merged Headers:** "Stem" is merged over two columns. Change to flat headers: `Stem_Height` and `Stem_Width`.
- **Blank Identifiers:** Location "Pahang" and Plant "Wild-01" are physically blank in rows 2 & 3. Row 2 looks like a floating leaf length (6.1) with no plant attached.
- **Two values in one cell:** "5.8 - 6.0" is text. No software can calculate the average leaf length of this column now.
- **Inconsistent missing data:** For Wild-02 Stem Width, a dash "-" is used. It should be left completely blank so the column remains strictly numeric.

How It Should Look (AI-Ready)

Location	Plant_ID	Stem_Height_cm	Stem_Width_cm	Leaf_Length_cm	Notes
Pahang	Wild-01	15.2	2.1	5.5	
Pahang	**Wild-01**	**15.2**	**2.1**	6.1	ID repeated
Pahang	**Wild-01**	**15.2**	**2.1**	**5.8**	Split from range
Pahang	**Wild-01**	**15.2**	**2.1**	**6.0**	Split from range
Pahang	**Wild-02**	14.5		4.2	Dead

**Highlighted cells show corrections: IDs filled down, ranges split into raw individual records, and missing data cleanly blank.*

Your Checklist for the Field

Determine the Row

Before making a sheet, ask: "What is ONE row?" Is it one plant? One leaf? Keep sheets separated by unit of observation.

Flat Headers

Exactly ONE header row per sheet. Never merge data-entry cells or header rows. Combine words with underscores.

Repeat IDs

Repeat Plant IDs, Dates, and Locations on every single row they apply to. Never rely on visual blanks for grouping.

Questions?

Thank you for your attention.

Image Sources

https://physicsworld.com/wp-content/uploads/2023/01/2023-01-Green-students-stressed-overworked-exhausted-1302022311-iStock_Flashvector.jpg

Source: physicsworld.com

Thumbnail for lacasadellafibra.com

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Source: lacasadellafibra.com